



FISHES of SAHUL



VOLUME THIRTY SEVEN
NUMBER TWO

DECEMBER 2023

JOURNAL OF THE AUSTRALIA NEW GUINEA FISHES ASSOCIATION

Incorporated Registration No. A0027788J. ISSN: 0813-3778 (print) ISSN: 2205-9342 (online)



Melanotaenia jakora, a new rainbowfish species from Papua Barat Province, Indonesia.

Photo: Erwin Binner

West Papua 2023 - experiences and results

Christophe Mailliet, Johannes Graf & Wim Heemskerk

2032

Salt, your next best friend

Peter J. Unmack

2058

Why salt helps Sahul fishes: osmosis explained

Rob Wager

2061

A new species of rainbowfish (Melanotaeniidae), *Melanotaenia jakora*, from Western New Guinea (Papua Barat Province, Indonesia)

Johannes A. Graf, Henderite L. Ohee, Fabian Herder & Haryono Haryono

2063

The discovery of *Melanotaenia jakora* and hints for keeping and breeding

Johannes A. Graf

2069

Fish in Focus: Drysdale Gudgeon *Hypseleotris notata* (Eleotridae)

Peter J. Unmack & Christopher Scharpf

2073

A new species of rainbowfish (Melanotaeniidae), *Melanotaenia jakora*, from Western New Guinea (Papua Barat Province, Indonesia)

Johannes A. Graf^{1*}, Henderite L. Ohee², Fabian Herder¹ & Haryono Haryono³

¹Museum Koenig, Leibniz Institute for the Analysis of Biodiversity Change, Bonn, Germany.

²Biology Department of Mathematics and Natural Sciences Faculty of Cenderawasih University, Waena Campus, Kamp. Woker Street, Waena 99358, Jayapura, Papua, Indonesia.

³Museum Zoologicum Bogoriense, Research Center for Biosystematics and Evolution, National Research and Innovation Agency (BRIN), Jl Raya Jakarta-Bogor, Km 46, Cibinong 16911, Indonesia.

*Corresponding Author: j.graf@web.de

<http://zoobank.org/urn:lsid:zoobank.org:pub:C9D08DD2-5F11-4BB1-93C4-54642F97F1F0>

Abstract

A new species of rainbowfish, *Melanotaenia jakora*, is described on the basis of 16 specimens, 34–63 mm SL from a rainforest creek near Meyado village in the Sebjar River system in the southern part of Vogelkop Peninsula of Papua Barat Province, Indonesia. The new species differs from all other *Melanotaenia* of the Vogelkop clade in the distinct combination of IV (III–V) + I, 10 (I, 9–12) dorsal fin rays, I, 20 (I, 18–22) anal fin rays, 14 (11–17) total gill rakers on the first arch, males with elongated dorsal and anal fins reaching base of caudal fin, and coloration of adult males lacking dark scale margins, showing a reflective blue shine of the body, and a solid midlateral stripe without disruptions.

Introduction

Rainbowfishes of the family Melanotaeniidae show a rich diversity in male coloration and pronounced patterns of local endemism Allen (1980). A total of 112 species are known from freshwater habitats in New Guinea and Australia and new species are continually being discovered, the latest by Hammer et al. (2019);

additional species are known but remain formally undescribed (Unmack et al. 2013).

Phylogenetic reconstructions suggests that the current taxonomy does not reflect monophyletic lineages in all cases, including the most species-rich genera *Melanotaenia*, *Chilatherina*, and *Glossolepis* (Unmack et al. 2013, Stelbrink et al. 2014). Essentially, the diversification of rainbowfishes is correlated with three main biogeographic regions: western New Guinea, northern New Guinea, and southern New Guinea plus Australia (Unmack et al. 2013).

Their attractive appearance makes rainbowfishes popular among aquarium hobbyists all over the world (Tappin 2011). However, many populations, especially in New Guinea, are under increasing pressure due to habitat loss (Jong 2021, Nugraha et al. 2015). This paper describes a new species of rainbowfish collected in the vicinity of Meyado village in the Sebjar River system (Fig. 1), southern Vogelkop Peninsula, and is apparently endemic to that area.

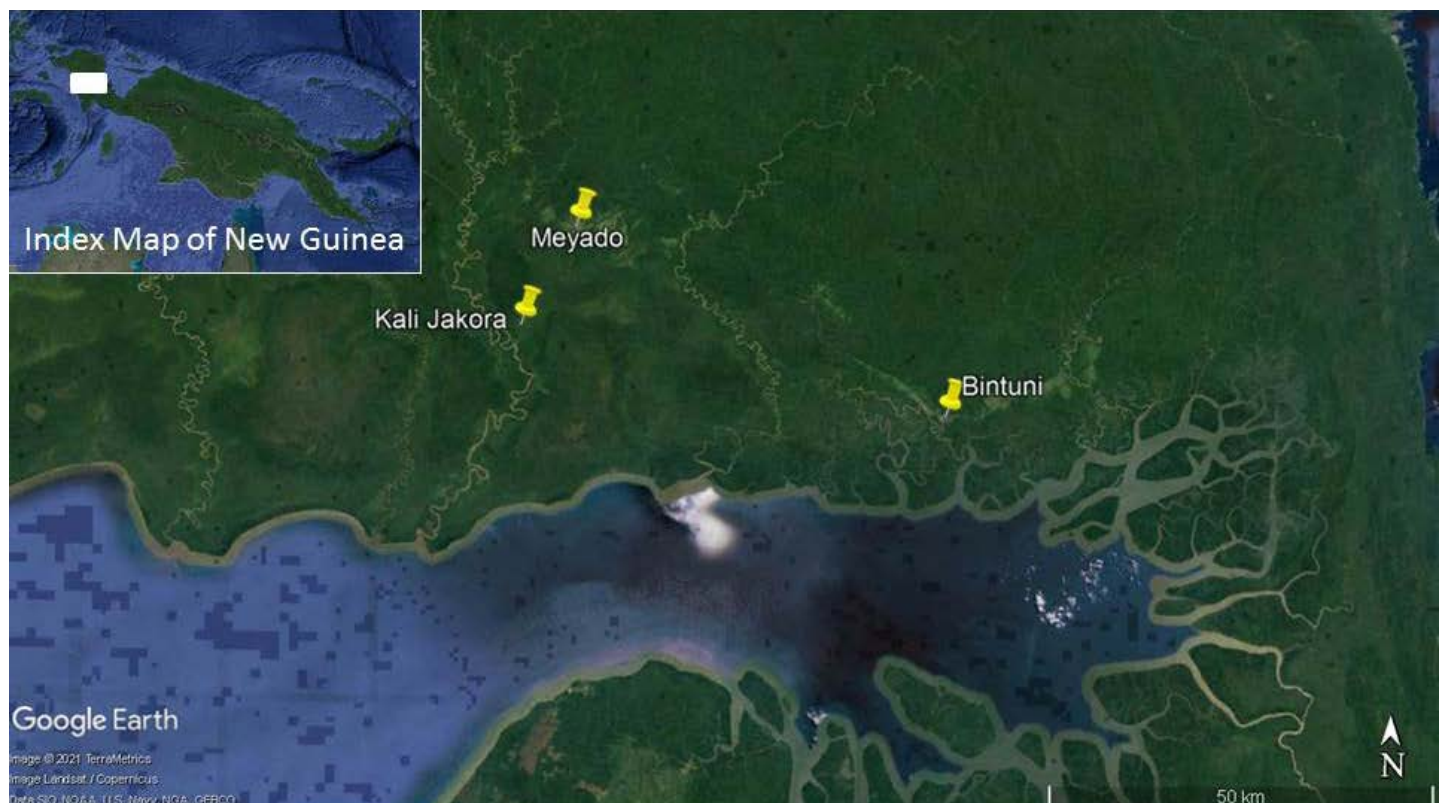


Figure 1. Map showing the type locality of *Melanotaenia jakora*, new species. Type locality indicated as “Kali Jakora”. (© map Wikimedia commons, Johannes Graf).



Figure 2. *Melanotaenia jakora*, new species, captive male specimens.

Photo: Erwin Binner

Material and Methods

Fishes were collected with hand nets on 11 August 2017 and were preserved in 70% ethanol. Photographs were taken in the field and in the aquarium to document live coloration. Counts and measurements that appear in parentheses refer to the range for paratypes if different from the holotype. Specimens are deposited at the Museum Zoologicum Bogoriense, Cibinong, Indonesia (MZB), the Museum and Arts Gallery of the Northern Territory, Darwin, Australia (NTM), and the Museum Koenig, Leibniz Institute for the Analysis of Biodiversity Change, Bonn, Germany (ZFMK).

Comparative material examined: *Melanotaenia ogilbyi* - ZMA 103110 (Lectotype), ZMA 103111 (2 specimens), 51–57 mm SL; *Melanotaenia sexlineata*, AMS I.20726-001 (holotype), AMS IA.7249 (3 specimens), AMS IA.7246 (1 specimen), AMS IA.7278 (2 specimens), 55–73 mm SL; *Melanotaenia garylangei* - MZB 22255 (Holotype) and MZB 22256 (5 specimens), 42–78 mm SL; ZFMK 68042-68047 (6 specimens), 42–63 mm SL; WAM P.34291-001 (3 specimens), 49–67 mm SL. Museum abbreviations used follow <https://researcharchive.calacademy.org/research/ichthyology/catalog/collections.asp>

The methods of counting and measuring follow Graf et al. (2015).

Species Account

Melanotaenia jakora sp. nov.

Jakora Rainbowfish

(Figures 2–5, Tables 1–2)

Holotype. MZB 26058, male, 63 mm SL, unnamed rainforest creek in the vicinity of Meyado, Sebjar River system, -2.02663639, 133.05784961, Papua Barat Province, Indonesia, depth 0.2 m, hand net, H. Ohee, W. Heemskerk, G. Lange and J. Graf, 11 August 2017.

Paratypes. MZB 26059, 5 specimens, 2 males and 3 females, 38–51 mm SL, collected with holotype.

NTM S.18674-001, 5 specimens, 2 males and 3 females, 40–54 mm SL, collected with holotype.

ZFMK ICH-128000-128004, 5 specimens, 3 males and 2 females, 34–59 mm SL, collected with holotype.

Diagnosis. A rainbowfish distinguished from all other *Melanotaenia* species of the Vogelkop clade I (Kadurusman *et al.* 2012a) by the following combination of characters: dorsal fin rays IV (III–V) + I,10 (9–12); anal fin rays I,20 (18–22); gill rakers on lower limb of first arch 14 (11–17); dorsal and anal fins elongated in adult males, tips reach base of caudal fin; live color differences of males to male *Melanotaenia* species of the Vogelkop clade: no dark margins of the scales, reflective blue shine of the body, solid midlateral stripe without color disruptions.

Description. Dorsal fin rays IV (III–V) + I,10 (9–12); anal fin rays I,20 (18–22); pectoral fin rays 13 (12–13); pelvic fin rays I,5 (I,5); branched caudal fin rays 15; lateral scales 37 (35–39); transverse scales 12 (10–13); predorsal scales 21 (18–25); cheek scales 11 (8–15); gill rakers on first branchial arch 14 (11–17). Body depth 3.4 (2.7–4.7) in SL; head length 4.3 (3.6–5.4) in SL; greatest width of body 2.8 (2.4–3.3) in greatest body depth; snout length 3.8 (2.8–4.4) in HL; interorbital width in head length 2.5 (2.3–2.7); eye diameter in head length 2.4 (2.2–2.7); caudal peduncle depth in head length 2.4 (2.0–2.7); caudal peduncle length in head length 1.4 (1.2–1.7). Jaws about equal, oblique, premaxilla without an abrupt bend between the anterior horizontal portion and lateral part; maxilla ends opposite or slightly posterior to front border of eye; lips thin, teeth conical with slightly curved tips; extending to outer surface of lips, teeth of upper jaw in 2 to 3 irregular rows anteriorly, reduced to a single row posteriorly where they are exposed when mouth is closed; teeth in lower

	MZB 26058	NTM S.18674-001	ZFMK ICH- 128000	NTM S.18674-001	MZB 26059	MZB 26059
	male	male	male	female	female	female
Standard length (mm)	63.00	54.00	45.00	41.00	41.00	38.00
Snout length	6.03	8.89	6.44	6.10	8.29	6.32
Head length	21.59	24.63	22.22	23.66	28.05	25.53
Interorbital width	9.52	9.07	9.33	9.51	11.71	10.53
Eye diameter	9.05	9.44	9.56	10.00	12.20	11.84
Body depth	36.03	34.44	30.22	26.83	36.59	25.79
Body width	10.95	12.59	10.00	10.00	13.41	10.53
Caudal peduncle depth	10.79	9.63	9.78	9.02	11.46	10.26
Caudal peduncle length	15.56	14.26	15.78	19.27	20.98	17.11
Predorsal length	43.81	50.00	46.44	47.56	55.61	47.63
Prepelvic length	31.11	33.70	34.89	33.90	43.90	33.95
Preanal length	41.43	45.74	45.33	46.83	59.02	50.00
Pectoral fin length	17.14	16.67	16.89	17.32	21.22	18.95
Ventral fin length	16.67	19.07	15.33	12.44	16.34	12.11
Longest spine of first dorsal fin	10.16	10.56	10.89	10.00	14.39	12.11
Longest spine of second dorsal fin	8.10	8.33	8.67	12.20	15.61	14.21
Spine length of anal fin	6.83	6.11	6.89	7.32	8.29	9.47
Total dorsal fin length	38.10	36.67	35.33	33.17	43.17	33.16
Second dorsal fin length	22.22	21.30	19.56	15.85	23.66	18.42
Anal fin length	42.70	43.33	37.56	32.93	42.68	33.16

Table I: Proportional measurements of selected specimens of *Melanotaenia jakora* expressed as percentage of the standard length (SL)

jaw in 5–6 irregular rows, reduced to 1 or 2 rows posteriorly, palatines edentate. Vomerine teeth small and conical. Scales of body large and arranged in regular horizontal rows; scale margins not crenulated; predorsal scales extending to posterior portion of interorbital; preopercle with one scale row between posterior angle to edge of eye. Predorsal length 2.2 (1.8–2.7) in SL; prepelvic length 3.0 (2.3–3.5) in SL; preanal length 2.2 (1.7–2.6) in SL; total dorsal fin length 2.9 (2.3–4.0) in SL; second dorsal fin length 5.3 (4.2–7.3) in SL; anal fin length 2.8 (2.3–4.0) in SL. First dorsal fin

originates opposite anal opening, first dorsal spine 2.3 (1.9–2.8) in HL; longest spines (2nd or 3rd) of males extending beyond origin of second dorsal fin when depressed; first spine of second dorsal fin 2.3 (1.7–3.0) in HL; first spine of anal fin 3.4 (2.6–4.4) in HL; pectoral fin length 1.4 (1.3–1.5) in HL; ventral fin length 1.7 (1.3–2.1) in HL.

Dorsal and anal fins of adult males extend to posterior caudal peduncle.



Figure 2. *Melanotaenia jakora*, new species, captive female specimen.

Photo: Erwin Binner

Colour in life (Fig. 2 and 3). Adult males with a blackish brown back, fading to a brown dorsolateral body, and a dark stripe covering two rows of scales. Below the dark stripe a whitish-silvery stripe extends along the lateral body, fading ventrally to blackish. Lower part of the opercle red. Both dorsal fins, anal and caudal fin with bluish white margins. Nape and first two membranes between spines of first dorsal fin and basis of second dorsal and anal fin reflective blue. Overall reflective blue coloration over the body in direct light. Central caudal fin with red spots, distal half of ventral fins red. Females brown with dark midlateral stripe of two scale rows and two faint orange lateral stripes in the ventral body. Faint orange coloration at the basis of caudal, second dorsal and anal fins.

Colour in alcohol (Fig. 4 and 5). Males with dark brown back; midlateral blackish stripe covering two scale rows; below the midlateral row a row of two scales with whitish color and faint dark lines between horizontal scale rows, below this line fading to blackish. Fins with whitish margins, ventral fins yellowish-hyaline. Second dorsal and anal fins dusky blackish. Females with paler brown color on upper side; centre of scales with lighter color compared to males; blackish midlateral stripe less intense than in males; lower half of body light brown. All fins light brown.

Sexual dimorphism. Sexual maturity is reached at a size of approx. 15 mm SL and 6 months of age, based on aquarium observations.

Males develop a deeper body than females, and have moderately elongated unpaired fins, compared with shorter fins in females. Colour of the unpaired and caudal fins in males translucent with bluish white margins, those of females translucent. In males, the tip of the first dorsal fin is whitish-blue and the center of the caudal fin is red.

Comparisons. The new species is most similar in appearance to *Melanotaenia gracilis* Allen 1978 from the Drysdale and King Edward rivers in the eastern Kimberley region of far northern Australia. It differs in having more soft rays in the second dorsal fin (9–12 vs. 7–10) more horizontal scale rows (10–13 vs. 9–10) and more vertical scale rows (35–39 vs. 33–35).

It differs from *Melanotaenia pygmea* Allen 1978 from the Prince Regent River in the western Kimberley region of far north Australia in having fewer spines in the first dorsal fin (3–5 vs. 5–7), more anal soft rays (18–22 vs. 14–18), more horizontal scale rows (10–13 vs. 8–9) and more vertical scale rows (35–39 vs. 31–33). In contrast to *Melanotaenia exquisita* Allen 1978 from King George River and upland tributaries of Katherine, Edith, Mary, South Alligator and Fergusson rivers in Western Australia and the Northern Territory, the new species has more soft rays in the second dorsal fin (9–12 vs. 7–9), more horizontal scale rows (10–13 vs. 9–10) and more lateral scale rows (35–39 vs. 34–35).

	Median	Min	Max	Holotype
First dorsal spines	4	3	5	4
Segmented second dorsal soft rays	10	9	12	11
Segmented anal soft rays	20	18	22	21
Pectoral rays	13	12	13	13
Segmented pelvic soft rays	5	5	5	5
Branched caudal rays	15	15	15	15
Lateral scales	37	35	39	38
Transverse scales	12	10	13	12
Predorsal scales	21	18	25	20
Cheek scales	11	8	15	12
Total gillrakers on first arch	14	11	17	14
[mm]	Mean	Min	Max	Holotype
Standard length (SL)		34.0	63.0	63.0
Snout length	2.8	2.0	4.8	3.8
Head length	10.6	8.3	13.6	13.6
Interorbital width	4.1	2.9	6.0	6.0
Eye diameter	4.3	3.3	5.7	5.7
Body depth	13.7	8.7	22.7	22.7
Body width	4.8	3.5	6.9	6.9
Caudal peduncle depth	4.4	2.9	6.8	6.8
Caudal peduncle length	7.4	6.2	9.8	9.8
Predorsal length	20.9	16.1	27.6	27.6
Prepelvic length	15.1	11.8	19.6	19.6
Preanal length	20.8	15.3	26.1	26.1
Pectoral fin length	7.8	6.2	10.8	10.8
Ventral fin length	6.5	4.1	10.5	10.5
Spine length of first dorsal fin	4.7	3.7	6.4	6.4
Spine length of second dorsal fin	4.8	3.8	6.4	5.1
Spine length of anal fin	3.1	2.4	4.3	4.3
Total dorsal fin length	15.9	10.7	24.0	24.0
Second dorsal fin length	8.9	6.4	14.0	14.0
Anal fin length	16.6	11.8	26.9	26.9

Table II: Summary tables of counts and meristic data of *Melanotaenia jakora*



Figure 4. *Melanotaenia jakora*, new species, photograph of MZB 26058, male holotype preserved in 70% ethanol.

Photo: Johannes Graf.



Figure 5. *Melanotaenia jakora*, new species, photograph of NTM S.18674-001, female paratype, preserved in 70% ethanol.

Photo: Johannes Graf

Melanotaenia jakora differs from all known rainbowfishes from the biogeographic zone Clade I of Vogelkop Peninsula (Kadarusman 2012) by its small (less than 65 mm SL) and slender (Body depth in SL 3.4 in average) body, presence of bluish white margins on the caudal fin, absence of black margins on the scales and absence of a black caudal fin margin.

Comparisons with *Melanotaenia sexlineata* (Munro 1964), *M. ogilbyi* Weber 1910, and *M. garylangei* Graf, Herder & Hadiaty 2015 show no strong similarity to species of the *maccullochi* group as referred to in Allen (1981) and Graf et al. (2015). *Melanotaenia jakora* has fewer spines in the first dorsal fin (3–5 vs. 5–7, 5–6 and 4–6, respectively), more anal soft rays (18–22 vs. 16–19, 17–18, 16–20, respectively), more lateral scales (35–39 vs. 31–33, 34–35, 34–37, respectively) and more predorsal scales (18–25 vs. 14–16, 15–16, 11–16, respectively).

Genetic analyses based on single nucleotide polymorphisms (SNPs) of nearly all described species of rainbowfishes and several undescribed populations, performed by P. Unmack at the University of Canberra, show that *Melanotaenia jakora* represents a lineage separated from all other rainbowfishes, being most closely related to species in the Birds Head lineage (Unmack 2023, pers. comm.).

Zoogeography and habitat. The new species was found in a small rainforest creek crossing the road from Meyado to the village of Jakora. The water at the collection site (Fig. 6) was dark with tannin. Typical habitats were characterized by muddy substrate; no submerged aquatic vegetation was present. *Melanotaenia jakora* was recorded in undisturbed habitats, partially covered by forest canopy, and disturbed sites open to sunlight. Water parameters were 26.4 °C, 30 uS, pH 6.4, hardness below measurability. The only fish species co-occurring with *Melanotaenia jakora* was *Melanotaenia* sp., a typical species of Clade I as mentioned in Kadarusman (2012).

Remarks. *Melanotaenia jakora* is already introduced into the aquarium hobby under the name “*Melanotaenia* sp. “Kali Jakora”. It reproduces well in soft, slightly acid (pH 6.5) water using a mop of acrylic yarn as egg-deposition substrate. Eggs are clear after deposition and turn light brown during development. Depending on water temperature, eggs need 8–10 days to hatch. Freshly hatched larvae feed on infusoria. In captivity, sexual maturity of the young fish can be reached in about 6 months.

Etymology. The new species is named after the village of Jakora, the closest named locality.



Figure 6. Type locality of *Melanotaenia jakora*, new species, in the vicinity of Jakora village, Sebjar River River system, Papua Barat Province, Indonesia.

Photo Henderite L. Ohee

Acknowledgements. Thanks are due to Peter Unmack, University of Canberra, for providing genetic data of the new species and Erwin Binner, Schwandorf, for providing pictures of the new species.

References

- Allen GR (1978) The rainbowfishes of northwestern Australia (family Melanotaeniidae). *Tropical Fish Hobbyist* 26: 91–102.
- Allen GR (1980) A generic classification of the rainbowfishes (Family Melanotaeniidae). *Records of the Western Australian Museum* 8: 449–490.
- Allen GR (1981) The “Maccullochi species group” of rainbowfishes (Melanotaeniidae) with the description of *Melanotaenia papuae*, new species. *Revue française d'Aquariologie Herpétologie* 8: 47–56.
- Graf JA, Herder F, Hadiaty RK (2015) A new species of rainbowfish (Melanotaeniidae), *Melanotaenia garylangei*, from Western new Guinea (Papua Province, Indonesia). *Fishes of Sahul* 29: 870–881.
- Hammer MP, Allen GR, Martin KC, Adams M, Unmack PJ (2019) Two new species of dwarf rainbowfishes (Atheriniformes: Melanotaeniidae) from northern Australia and southern New Guinea. *Zootaxa* 4701: 201–234.
- Jong HN (2021) Papua deforestation highlights eastward shift of Indonesia forest clearing. <https://news.mongabay.com/2021/03/papua-deforestation-highlights-eastward-shift-of-indonesia-forest-clearing>
- Kadariusman, Hubert N, Hadiaty RK, Sudarto, Paradis E, Pouyaud L (2012) Cryptic diversity in Indo-Australian rainbowfishes revealed by DNA barcoding: implications for conservation in a biodiversity hotspot candidate. *Plos One* 7: 1–8.
- Nugraha MFI, Pouyaud L, Carman O, Widyastuti U, Junior MZ, Kadariusman, Avarre J (2015) Genetic diversity of Boeseman’s rainbowfish (*Melanotaenia boesemani*) reared in Indonesian farms compared to endangered natural populations. *Tropical Conservation Science* 8: 796–812.
- Stelbrink B, Stöger I, Hadiaty RK, Schliewen UK, Herder F (2014) Age estimates for an adaptive lake fish radiation, its mitochondrial introgression, and an unexpected sister group: sailfin silversides of the Malili Lakes system in Sulawesi. *BMC Evolutionary Biology* 14: 94.
- Tappin A (2011) Rainbowfishes ~ Their Care & Keeping in Captivity Second Edition. <http://www.mediafire.com/download/g7qzn85uqde8v8o/Rainbowfishes.2011.pdf>
- Unmack PJ, Allen GR, Johnson JB (2013) Phylogeny and biogeography of rainbowfishes (Melanotaeniidae) from Australia. *Molecular Phylogenetics and Evolution* 67: 15–27.

